

| Question number | Answer                                                                 | Mark |
|-----------------|------------------------------------------------------------------------|------|
| 1(a)(i)         | D the hydrophobic tails move away from the aqueous (water) environment | (1)  |

| Question number | Answer | Mark |
|-----------------|--------|------|
| 1(a)(ii)        | B Q    | (1)  |

| Question number | Answer                                                                                                                                                | Mark |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(b)            | <ul style="list-style-type: none"> <li>diffusion of water molecules down a water potential gradient through a partially permeable membrane</li> </ul> | (1)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                               | Additional guidance                    | Mark |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------|
| 1(c)            | <p>An answer that includes the following points:</p> <p>similarities:</p> <ul style="list-style-type: none"> <li>both used to transport large particles / large quantities of material (1)</li> <li>both involve (phospholipid) membrane vesicles (1)</li> </ul> <p>difference:</p> <ul style="list-style-type: none"> <li>exocytosis is export and endocytosis is import of material (1)</li> </ul> | Accept a description of both processes | (3)  |

| Question number | Answer                                                                                                                                                                                                                                                                               | Mark |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1(d)            | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>oxygen molecule is small (1)</li> <li>oxygen molecule is non-polar (1)</li> <li>(it can, therefore) pass between gaps {in cell membrane / between hydrophobic tails} (1)</li> </ul> | (3)  |